

Explain how an inversion mutation can cause the production of non-functional factor VIII.

[illegible]

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The scientists:

- selected 112 male volunteers with severe HA, who were at least 18 years of age
- treated each volunteer with a single injection of Roctavian into their blood
- recorded the number of bleeding episodes (internal bleeding) in these volunteers over a two-year period (set **A** results)
- compared these data with the number of bleeding episodes per year in the volunteers **before** receiving Roctavian (set **B** results).

Table 1

Volunteer set	Mean number of bleeding episodes per year ($\pm 2 \times \text{SD}$)
A (Received Roctavian)	0.8 $\pm$ (1.2)
B (Received factor VIII)	4.8 $\pm$ (0.6)

Use all the information provided to evaluate the use of Roctavian as a treatment for HA.

[5 marks]

[illegible]

Question 4 continues on the next page

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Haemophilia C is an **autosomal** recessive condition.

Two parents are both heterozygous for the haemophilia C allele.

Calculate the probability of these parents producing a son who has haemophilia C.

[1 mark]

Probability = _____

1

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